Prediction Markets

MGT 4074 Fintech & Crypto Tokens, Fall 2025

Joseph Hall

Georgia Institute of Technology

October 22, 2025

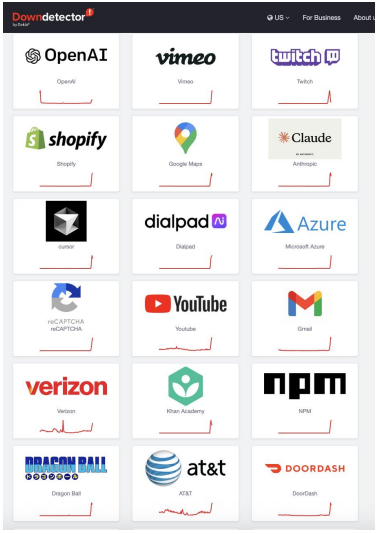
Roadmap

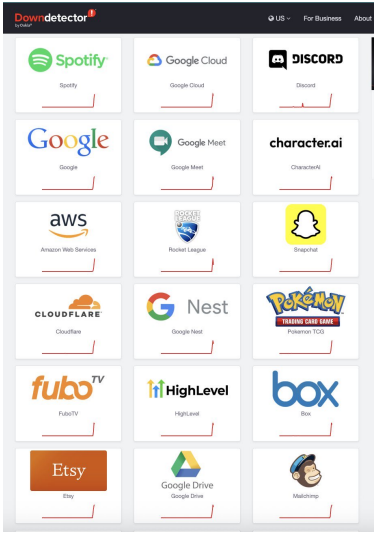
1. In the News
2. Prediction Platforms
3. Gambler's Math
4. Decentralized Prediction Markets

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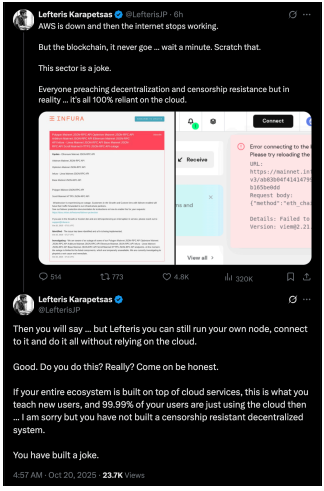
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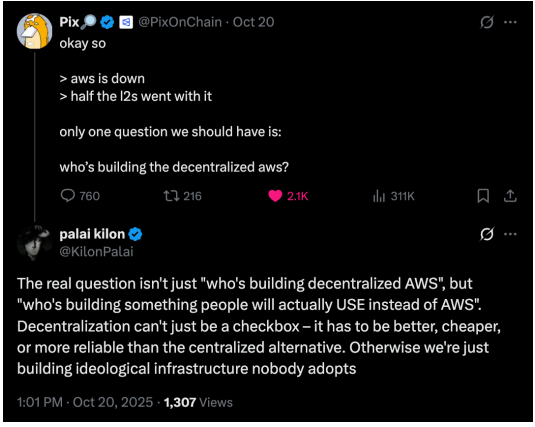
AWS Outage Breaks the Internet





Impact of Cloud Service Outage on DeFi





Roadmap

1 In the News

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Definition

In a **Prediction Market**, bets are traded on binary outcomes: $1 if prediction comes true, $0 if prediction does not.

There are two major market structures of prediction markets:

- In **Proprietary** markets, one entity ("the house") lists odds unilaterally and takes the opposite side of every customer's bet.
  - These are often called "sportsbooks", "bookies", "oddsmakers" or just "Vegas".
  - The prices offered for complementary predictions add up to much more than $1 - there is a "house edge" which is called a "vig" or "juice"
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  - The house usually tries to match its book (sell equal quantities of yes and no) – "the house always wins"
- In **Peer to Peer** markets, opposite sides of the bet must match exactly and agree on a market price to create a bet.
  - Kalshi uses a P2P structure despite being a centralized platform; it takes a transparent fee instead of a house edge

Behavior in Prediction Markets

From Matt Brown Econ:

- The average sports better is *highly overconfident:* They expect to break even (and enjoy the variance) but in fact lose 7.5 cents on the dollar:
- Overoptimism is worse among bettors who prefer **parlays**: compound bets that all of several (usually uncorrelated) events ("legs" of the parlay) will all occur successfully with no failures
- Summarizing bettors' historical performance does nothing to correct their bias, but allowing them to set limits in advance reduces gambling by a modest amount

Worldwide, about 60% of gambling losses (aka gambling industry profits) come from gambling addicts per the World Health Organization.

Popular Prediction Markets

Popular prediction markets include:

- **Kalshi**, a centralized and regulated for-profit all-purpose prediction market
- **Polymarket**, a decentralized and unregulated all-purpose prediction market
- **Predictit**, a centralized and regulated non-profit prediction market focused on political outcomes
  - I dabbled in Predictit in college, but it was impossible to make money after fees
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Economists love prediction markets for the same reason we love stock markets: they incentivize people to *gather and reveal information* that the rest of us get *for free!*

Polymarket Growth

Polymarket Monthly Volume

- Since Polymarket is on-chain, Dune can make an easy dashboard!
- Dune uses Structured Query Language (SQL), which you should learn if you haven't already (it's incredibly easy)
- TPan on Substack

Illegal Event Categories

A prediction is a type of derivative. Many derivative markets are *per se*[1] illegal!

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  - A popular financial product was once the **Tontine**: a group pools funding, and each receives annuity payouts until they die. Payouts increase as one outlive their peers
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- **Motion Picture Box Office Receipts**: Swept up in 2008 Dodd-Frank (story)
- **Onion Futures**, banned by Congress in 1958 in response to the Great Onion Corner of 1955 where one man came to own every onion in Chicago and was able to extort wholesalers

Prediction markets are not regulated by the SEC, but rather the **Commodities Futures Trading Commission (CFTC)**. Predictions (about things that aren't securities) are considered "Arrow Securities" by academics, but ironically are not considered "securities" by U.S. law! (who remembers the *Howey Test*?)

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CFTC Signals Support for Decentralized Markets

Innovation Exemptions and Decentralized Finance

Today's decentralized finance (DeFi) protocols enable direct peer-to-peer trading without the need for intermediaries. We reaffirm that both agencies are prepared to consider "innovation exemptions" to create safe harbors or exemptions that allow market participants to engage in peer-to-peer trading of spot, leveraged, margined, or other transactions in spot crypto assets, including derivatives such as perpetual contracts, over DeFi protocols. These safe harbors and exemptions would allow market participants to build commercially viable models while the agencies advance longer-term rulemaking.

The right to self-custody one's assets is a core American value. While market participants have paths under current law to trade spot crypto on federally regulated venues, the path remains open for peer-to-peer spot crypto trading as well. We encourage market participants to meet with our respective staffs as entrepreneurs onshore trading activity and innovate.

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Sports books list prices using **Moneyline Odds**:

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  - Use the formula $\frac{X}{100+X}$ to convert odds of $-X$ to a price

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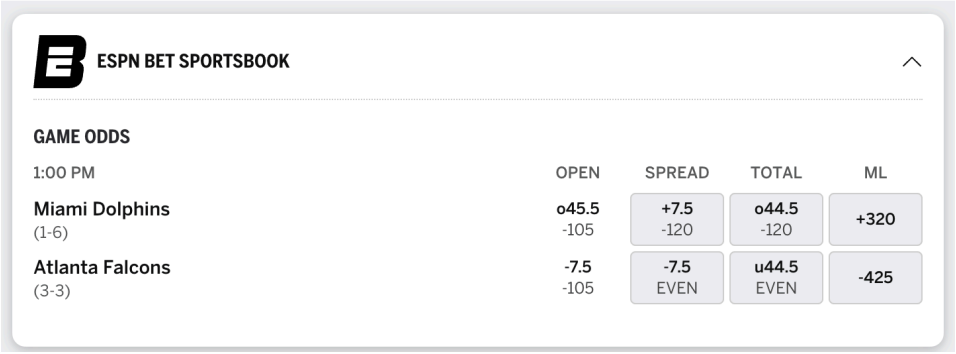
The **Moneyline** is contrasted with the **Spread**, which is an equal-odds bet on the margin of victory. When two teams are unevenly matched, the "line" can be "drawn" in probability space (the "moneyline") or in margin-of-victory space (the "spread")[3]

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Example: ATL vs. MIA, Oct 26 2025

What is the implied probability that the Atlanta Falcons defeat the Miami Dolphins?



ESPN BET SPORTSBOOK

GAME ODDS

1:00 PM	OPEN	SPREAD	TOTAL	ML
Miami Dolphins (1-6)	o45.5 -105	+7.5 -120	o44.5 -120	+320
Atlanta Falcons (3-3)	-7.5 -105	-7.5 EVEN	u44.5 EVEN	-425

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Q: Why do they list it this way?

A: To confuse you and hide the house edge.

Kelly Criterion

The **Kelly Criterion** is a formula, derived using basic statistics and calculus, to determine how large of a wager to make ("**bet sizing**")

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It maximizes expected wealth over many **independent (uncorrelated) bets**

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- **Time Value of Money** – the asset is traded today but the payoff of the bet might be a long time from now (e.g. 2028 Presidential Election), necessitating an adjustment for time value of money.
- **State Prices** – Predictions strongly correlated with macroeconomics can trade above or below probability because of **state prices**: a dollar in the state of the world where we have a recession becomes especially valuable. Therefore that prediction will trade above true probability.

The second term is not generally possible to calculate without strong assumptions. The first term is easy – just inflate the price by the time value (e.g. maturity-matched treasuries) between the trade date and the settlement date to get the probability. **For bets resolving in the near future and uncorrelated with the macroeconomy, prices in a frictionless competitive market should trade to true probability**

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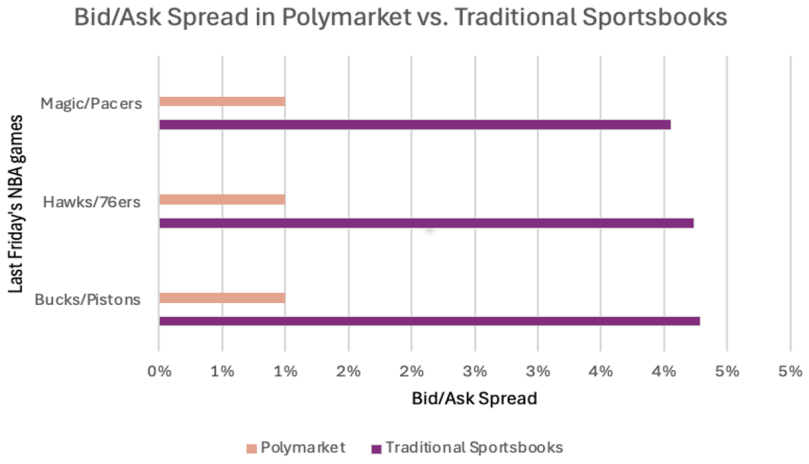
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Polymarket

Quick Overview

- On Polymarket, you can **buy and sell shares** representing future event outcomes (i.e. "Will TikTok be banned in the U.S. this year?")
- Shares in event outcomes are **always priced** between 0.00 and 1.00 **USDC**, and every pair of event outcomes (i.e. each pair of "YES" + "NO" shares) is fully collateralized by $1.00 USDC.
- Shares are created when **opposing sides come to an agreement on odds**, such that the sum of what each side is willing to pay is equal to $1.00.
- The shares representing the *correct, final outcome* are paid out $1.00 USDC each upon **market resolution**.
- Unlike sportsbooks, you are not betting against "the house" – the counterparty to each trade is another Polymarket user. As such:
  - Shares can be sold before the event outcome is known_ (i.e. to lock in profits or cut losses)
  - *There is no "house" to ban you for winning too much.*

Polymarket vs. Sportsbook "House Edge"



Source: Nils Bognar, MGT4074 Spring 2025 Final Project

Oracle Risk in Decentralized Markets

Just as in all other decentralized markets, decentralized prediction markets are heavily exposed to **Oracle Risk**: the chain has no eyes of its own and must be told what has occurred in the real world.
Polymarket uses the **UMA Optimistic Oracle** to resolve bets.

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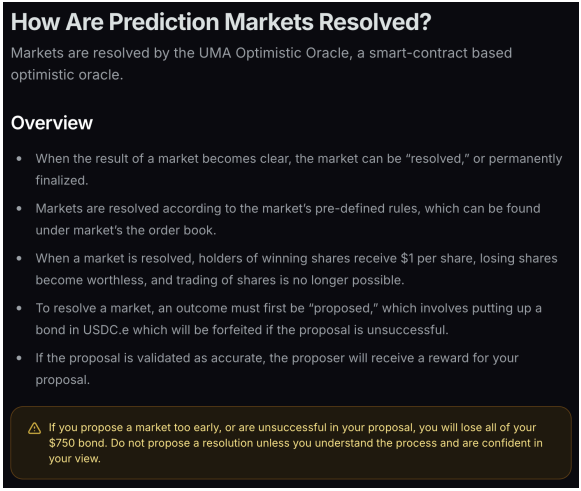
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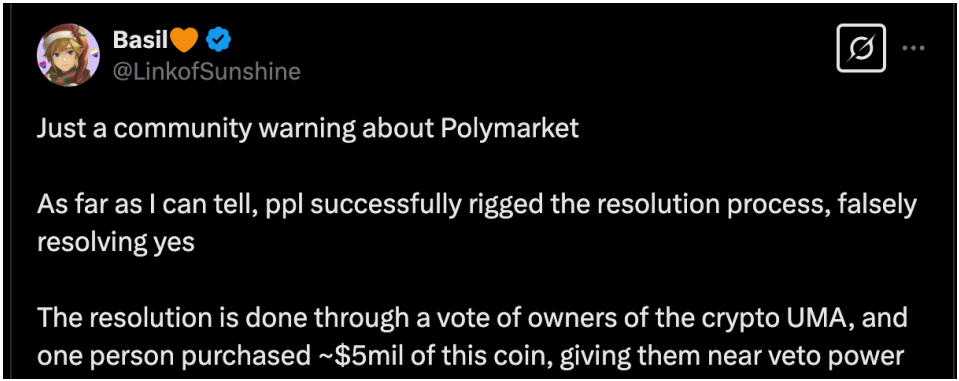
- "Optimistic" means that users put down collateral to propose to resolve outcomes. The collateral is lost if their proposed resolution is successfully challenged.
- UMA is the governance token for the optimistic oracle. *Owners of UMA tokens comprise the "jury" in the case of any dispute*

If you own a majority of UMA Tokens, you can rig the oracle! The creators of UMA call this "profit-from-corruption" or "PFC" risk in their whitepaper.

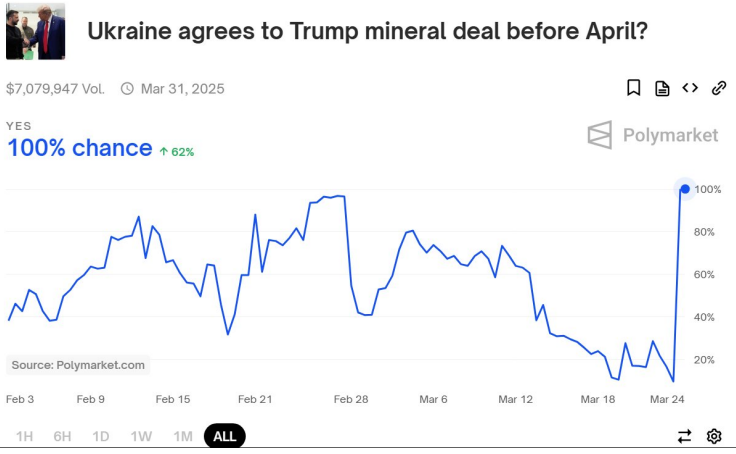
UMA Optimistic Oracle



Polymarket Oracle Failure



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